

Introduction

In the lead-up to a set of education reforms introduced in 2020, there was a perceived mismatch between popular fields of study and areas of job shortage in the Australian economy. The federal government's response was the Job-Ready Graduates Scheme (JRGS), implemented in 2021, which restructured domestic undergraduate tuition fees to incentivise high-priority disciplines (education, health, agriculture, engineering, and IT) through increased commonwealth funding and reduced student contributions, while discouraging low-priority disciplines such as management & commerce, law, arts, and social sciences through the inverse. Before the JRGS, maximum student contributions across disciplines ranged from roughly 6,500 to 11,000 per annum; following the reform, priority disciplines fell to approximately 4,000–9,500 while non-priority disciplines rose to 11,500–16,500, producing a spread of up to \$12,500 within a single year of study. Literature on tuition price elasticity suggests demand is inelastic at the market level but elastic when comparing variations across competing institutions (Carter & Curry, 2011), with other factors including location, curriculum, and interpersonal relationships also shaping enrolment decisions (Langelett et al., 2015). Existing evaluations of the JRGS have likened it to 'a misguided arrow missing its mark', arguing it is 'unlikely to have a significant effect on student choices' and that disciplinary selection is more heavily driven by socio-economic background, career guidance, and occupational expectations not addressed by the scheme (Daly & Lewis, 2020). However, these evaluations do not examine whether the policy may have succeeded selectively across specific disciplines, a gap this paper addresses by providing the first discipline-level evaluation of the JRGS using a cross-national difference-in-differences design.

This paper tests whether the JRGS produced enrolment shifts consistent with its intended pricing incentives, and whether any observed shifts align with the policy design. The approach uses a difference-in-differences estimator with UK and New Zealand (NZ) university enrolments as treatment-absent control groups. Both represent appropriate comparators: they share similar higher education structures, income-contingent loan schemes, and broadly comparable enrolment patterns, yet neither underwent a discipline-specific fee reform during the study period. The UK's domestic fee cap (£9,250 per annum) has been frozen uniformly across all subjects since 2017, meaning UK enrolment variation by discipline reflects demand and supply factors rather than price incentives. NZ's Fees Free policy (effective 2018) subsidised the first year of study universally across all disciplines, placing it fully in the pre-treatment window and rendering it orthogonal to discipline-level identification. University enrolment panel data spanning 2016–2024 across eleven disciplines for all three countries is used, alongside panel data on Australian university funding clusters and job shortage indicators. The estimating equation applies two-way fixed effects (country and year) with HC3 standard errors, justified by the small number of country clusters ($G=3$).

The findings suggest the JRGS had a statistically significant effect in two disciplines: education enrolments increased by approximately 18.6% ($p=0.035$) relative to control countries after 2021, while management & commerce fell by approximately 21.6% ($p=0.055$). The remaining nine disciplines showed no statistically significant change. While the two disciplines with the largest fee changes moved in the direction the JRGS intended, consistent with students responding to price signals, and these results alone cannot establish whether this reflects genuine demand elasticity or concurrent confounding factors. The null effects across the remaining disciplines suggest the scheme had limited broader impact in realigning enrolment with labour market priorities.

Literature Review

Price elasticity of demand for changes in tuition prices has historically been considered inelastic at the market level (Langelett, et al., 2015) and functionally 'close to zero' (Havranek, et al., 2018). Changes in overall pricing often lead to no significant change in enrolments, as regardless of tuition costs, most individuals seeking tertiary education will still pursue it. There are exceptions, such as between competing universities where price differences are suggested to elicit a response from prospective students when deciding where to study (Carter & Curry, 2011); however, this effect does not hold when market prices change uniformly, as with the JRGS. Instead, decisions to pursue tertiary education are suggested to be less dependent on pricing alone; factors such as availability, location, curriculum and discipline choice, and interpersonal relationships all contribute significantly to the decision-making process (Langelett, et al., 2015).

Applied to the JRGS, this literature suggests that market-level pricing changes are unlikely to affect overall enrolment, as the reform operated at a level where demand is inelastic. However, the specific targeting of high-priority and low-priority disciplines could potentially resemble the price elasticity observed between competing universities, where one institution is more expensive than another, but applied instead to disciplines competing within the same institution. This introduces its own complications. Literature surrounding university enrolment motivations is in tension with the use of price signals to direct disciplinary choice. Enrolment decisions are shaped by balancing intrinsic and extrinsic motivations; by radically shifting prices in either direction, the JRGS places a premium on financial costs as an extrinsic factor, which may crowd out intrinsic factors such as personal interest or enjoyment. Intrinsic motivation is important for sustaining long-term persistence in study, and where it is displaced, students are more likely to struggle with motivation, satisfaction, and completion of their degree (Janke, 2020). Utilising price signals to tilt the balance toward extrinsic factors penalises students for prioritising intrinsic motivations deemed 'low-priority'. If the JRGS does affect enrolment, it may consequently harm student welfare for those not intrinsically motivated toward the designated high-priority disciplines.

Specific literature evaluating the JRGS is limited, though the available findings are consistent with an inelastic demand model and are broadly critical of the policy. Prior pricing changes, including adjustments to HECS/HELP contribution rates, did not produce meaningful changes in enrolments despite their magnitude being smaller than those introduced by the JRGS (Daly & Lewis, 2020). This can be partially attributed to the structure of the HECS/HELP scheme itself, which permits students to defer fees until their income exceeds the repayment threshold, potentially reducing the perceived financial cost of a degree at the point of selection. It can also be attributed to the variation in personal and preference factors that are suggested to hold greater influence over tertiary enrolment decisions than price alone (Daly & Lewis, 2020).

Data and Methodology

Data

Data was collected across three countries: Australia (treatment), the United Kingdom (control), and New Zealand (control). Additional enrolment data was collected for Canada as a potential extra control, however it was ultimately excluded from the primary analysis as it introduced excess noise into the panel and was instead retained as an auxiliary robustness check. Panel data was utilised for all three countries.

Three categories of data were collected. First, university enrolments for all three countries spanning 2016–2024 across all disciplines and fields of study, which served as the centrepiece of the difference-in-differences analysis. Second, Australian job market shortage data across a range of industries, which was excluded from the primary econometric analysis and used instead to provide contextual support for the classification of disciplines as high-priority or low-priority under the JRGS. Third, tuition fees and commonwealth funding clusters for Australian tertiary disciplines across 2016–2024, which were matched to their respective disciplines to quantify how student contributions and government co-payments changed as a result of the reform.

Data Cleaning and Sample Restrictions

Disciplinary enrolments across the three countries were assigned common category keys to ensure each Australian treatment discipline had a directly comparable control counterpart in both the UK and New Zealand. The UK HESA data reported enrolments by academic year (e.g. 2016/2017), creating a six-month misalignment with the Australian and New Zealand calendar-year data; this was resolved by mapping each academic year to its start year (e.g. 2016/2017 became 2016), producing a consistent annual time index across all three countries. Australian funding clusters were also standardised across all time periods, as different versions of the raw data classified certain disciplines across different clusters; these were corrected to ensure that funding figures accurately reflected each discipline's actual allocation throughout the study window. The same category key system was then used to link funding clusters to their corresponding disciplines.

Certain disciplines were not consistently categorised in the raw data for the years 2016–2018, particularly for UK subject classifications which changed in 2019/2020. Observations for these disciplines prior to 2019 were dropped to preserve comparability, limiting five disciplines (natural and physical sciences, environment and related, society and culture, others, and architecture and building) to a 2019–2024 panel. The final observation count per discipline is N=27 (3 countries × 9 years) for disciplines with the full 2016–2024 panel, and N=18 (3 countries × 6 years) for disciplines restricted to 2019–2024. A log transformation was applied to the enrolment outcome variable to express results as percentage changes and to accommodate the substantial differences in nominal enrolment scale across the three countries. Together, these adjustments produce a balanced and comparable panel in which the primary source of structural variation is the JRGS shock rather than differences in national counting conventions.

Treatment Intensity by Discipline

Table D1. Maximum Student Contribution by Discipline — Pre- and Post-JRGS (AUD per EFTSL)

Discipline	JRGS	Pre-JRG (avg 2019–20)	2021	2024	Change (\$)	Change (%)
Management & Commerce	Non-priority	\$11,056	\$14,500	\$16,323	+3,444	+31.1%
Creative Arts	Non-priority	\$6,625	\$9,522	\$10,718	+2,897	+43.7%
Society & Culture	Non-priority	\$7,364	\$11,636	\$13,099	+4,273	+58.0%
Health	Priority	\$9,462	\$8,451	\$9,512	-1,011	-10.7%
Information Technology	Priority	\$9,443	\$7,950	\$8,948	-1,493	-15.8%
Engineering & Related Tech	Priority	\$9,443	\$7,950	\$8,948	-1,493	-15.8%
Architecture & Building	Priority	\$9,443	\$7,950	\$8,948	-1,493	-15.8%
Natural & Physical Science	Priority	\$9,487	\$7,518	\$8,461	-1,969	-20.8%
Education	Priority	\$6,625	\$3,950	\$4,445	-2,675	-40.4%
Environment & Related	Priority	\$9,443	\$5,213	\$5,867	-4,230	-44.8%

Source: Australian Government Department of Education, Annual Funding 2019–2026.

The JRGS produced a wide dispersion in fee changes across disciplines that had previously been relatively compressed. Before the reform, maximum student contributions ranged from approximately 6,500 to 11,000 per equivalent full-time student load. By 2021, that range had widened to roughly 4,000–15,000, with priority disciplines receiving cuts of between 20% and 46% and non-priority disciplines absorbing increases of 28–36%. Education received the largest proportional cut among priority fields, while management and commerce faced the steepest increase among non-priority fields; these are also the two disciplines that produce the only statistically significant results in the analysis. The variation in treatment intensity across disciplines provides a natural dose-response test: if fee signals drive enrolment decisions, the largest effects should be concentrated in the disciplines with the largest fee changes, which is broadly consistent with the findings.

Summary Statistics

Table D2. Summary Statistics — Enrolment Panel (AUS, UK, NZ; all disciplines)

Discipline	Panel	N	Mean enrolments	SD enrolments	Mean log(enrol)	SD log(enrol)	Pre-2021 mean log	Post-2021 mean log
Natural & Physical Science	Short (2019–2024)	30	79,874	39,617	11.162	0.519	11.171	11.157
Information Technology	Short (2019–2024)	18	106,953	68,546	11.225	1.003	11.133	11.270
Engineering & Related Tech	Full (2016–2024)	27	99,609	71,466	10.855	1.474	10.868	10.838
Architecture & Building	Full (2016–2024)	27	34,915	24,332	9.947	1.251	9.887	10.021
Environment & Related	Short (2019–2024)	30	17,850	9,169	9.639	0.593	9.621	9.647
Health	Full (2016–2024)	45	143,745	132,375	11.261	1.238	11.227	11.303
Education	Full (2016–2024)	27	93,750	58,845	11.026	1.123	11.034	11.016
Management & Commerce	Full (2016–2024)	27	295,223	196,631	12.171	1.110	12.139	12.210
Society & Culture	Short (2019–2024)	48	147,098	99,254	11.692	0.640	11.695	11.691
Creative Arts	Full (2016–2024)	27	101,065	66,649	11.223	0.868	11.235	11.208
Others	Short (2019–2024)	24	18,220	19,351	8.426	2.087	8.522	8.378

Units: enrolments = headcount; log(enrolments) = natural log of annual headcount. Total observations: 330 | Countries: AUS (treated), UK (control), NZ (control). Full panel (N=27 each): Engineering & Related Tech, Architecture & Building, Health, Education, Management & Commerce, Creative Arts. Short panel (N=18 each): Natural & Physical Science, Information Technology, Environment & Related, Society & Culture, Others.

Sample Construction and Exclusions

Table D0. Sample Construction

Step	Dropped	Remaining
Raw data collected	—	538
Restricted to 2016–2024 study window	79	459
Restricted to 11 matched category keys	45	414
Dropped pre-2019 for 5 UK-reclassified disciplines	84	330
Dropped rows with missing enrolment values	0	330

Final analytic sample: 330 observations. 6 disciplines × 3 countries × 9 years = 162 (full panel, N=27 each); 5 disciplines × 3 countries × 6 years = 90 (short panel, N=18 each).

Data Relation to Hypothesis

The hypothesis is that the JRGS fee changes shifted enrolments toward high-priority disciplines and away from those facing fee increases. The data structure is designed to isolate this effect: Australia is the only country in the panel to have altered its higher-education fee structure, while the UK and NZ held their funding arrangements effectively constant over the 2021–2024 period. The panel structure ensures that within-country, within-discipline variation over time drives the estimation rather than permanent scale differences between Australia and the control countries. The JRGS treatment also varies in intensity across disciplines, and disciplines that experienced larger fee movements are expected to exhibit larger enrolment responses if a treatment effect exists. Year fixed effects absorb any global shocks common to all three countries in a given year, most notably the disruptions associated with COVID-19, leaving only the Australia-specific post-2021 divergence in enrolments to be attributed to the JRGS.

Consistency of JRGS Fee Changes with Australian Job Shortages

Table D3. JRGS Fee Direction vs Australian Job Shortage Rates (Skill Level 1 Occupations)

Discipline	JRGS	Fee	2021 (%)	2023 (%)	2025 (%)	Avg 2021–25 (%)	Consistent?
Society & Culture	Non-priority	Increase	14.3	24.0	24.0	21.3	Partial / No
Management & Commerce	Non-priority	Increase	14.0	16.1	4.3	11.2	Yes
Creative Arts	Non-priority	Increase	0.0	0.0	2.9	0.6	Yes
Health	Priority	Decrease	10.4	80.5	74.4	63.1	Yes
Engineering & Related Tech	Priority	Decrease	30.3	76.5	47.1	54.3	Yes
Information Technology	Priority	Decrease	38.5	69.0	20.7	51.8	Yes
Architecture & Building	Priority	Decrease	0.0	60.0	100.0	44.0	Yes
Education	Priority	Decrease	5.9	47.4	47.4	38.0	Yes
Natural & Physical Science	Priority	Decrease	33.3	29.4	32.4	30.2	Yes
Environment & Related	Priority	Decrease	16.7	3.8	0.0	4.9	Partial / No

Source: Department of Employment and Workplace Relations, Occupation Shortage List 2021–2025. Skill Level 1 occupations require a bachelor degree or higher.

The JRGS fee structure is broadly consistent with the pattern of labour market shortages across the 2021–2025 period. Among the seven priority disciplines, health, engineering, education, and information technology all recorded persistently high shortage rates, peaking between 47% and 80% of Skill Level 1 occupations in national shortage at various points across the window. Among non-priority disciplines, management and commerce and creative arts show consistently low shortage rates (averaging under 15% and 1% respectively), supporting the rationale for the fee increases applied to those fields.

The alignment is not universal. Architecture and building, classified as a priority discipline, showed near-zero shortage rates until 2023 before rising sharply to 100% by 2025, a pattern more consistent with a lagged cyclical demand shock than a structural supply shortage the policy was calibrated against. Society and culture, a non-priority field subject to fee increases, has seen a gradual upward trend driven primarily by legal and social work occupations. Nevertheless, for the disciplines central to this paper's findings (education and health on the priority side; management and commerce on the non-priority side), the fee changes are consistent with the labour market evidence, lending contextual support to the policy's stated objectives.

Methodology

The estimating equation is a two-way fixed effects difference-in-differences model, estimated separately for each discipline:

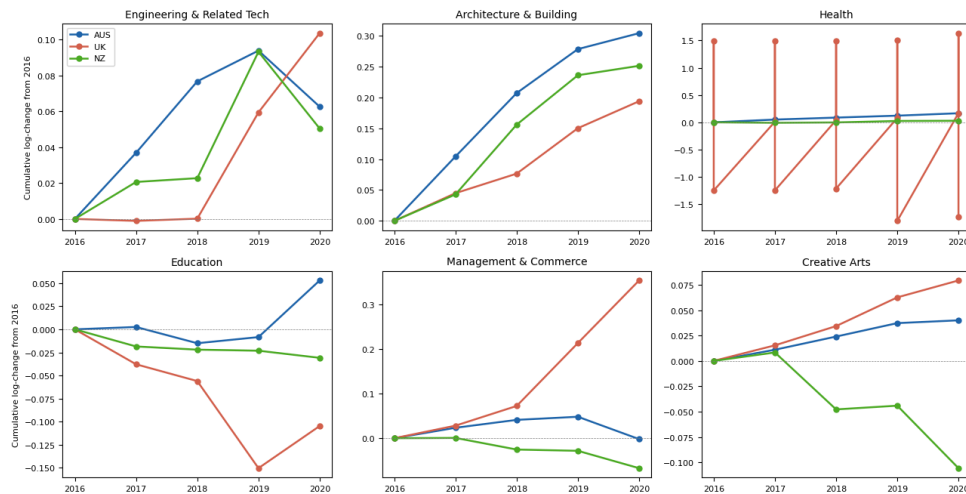
$$\log E_{ct} = \alpha + \beta_1 \text{treated}_c + \beta_2 \text{NZ}_c + \beta_3 (\text{treated}_c \times \text{post}_t) + \sum_{t=2017}^{2024} \gamma_t \mathbf{1}[t] + \varepsilon_{ct}$$

where $c \in \{\text{AUS, UK, NZ}\}$ and t indexes calendar years. treated_c and NZ_c are country indicators that absorb permanent level differences across the three countries, with the UK as the reference unit. $\text{post}_t = \mathbf{1}[t \geq 2021]$ marks the JRGS implementation period. Year fixed effects γ_t absorb shocks common to all three countries in a given year, including the global disruption from COVID-19. The coefficient of interest is $\hat{\beta}_3$, which identifies the post-2021 deviation of Australian log-enrolments from the counterfactual trajectory implied by the pooled UK–NZ trend. The approximate percentage effect is $(e^{\hat{\beta}_3} - 1) \times 100$.

Standard errors are HC3 throughout. With $G=3$ country clusters, cluster-robust inference is not feasible: the Moulton factor for this panel implies non-trivial within-cluster error correlation, and the minimum cluster count for asymptotically reliable cluster-robust SEs is approximately 42. HC3 provides a conservative alternative. The primary specification leaves 15 residual degrees of freedom for full-panel disciplines ($N=27$) and 9 for short-panel disciplines ($N=18$).

The identifying assumption is that, absent the JRGS, Australian enrolments in each discipline would have followed a trend parallel to the pooled UK and NZ counterfactual from 2021 onward. This is evaluated below using pre-treatment trends for the six full-panel disciplines, where five years of pre-treatment data are available.

Pre-treatment log-enrolment trends — full-panel disciplines, indexed to 2016 (AUS, UK, NZ)



Pre-treatment trends are broadly parallel for most full-panel disciplines, though with meaningful variation. Management and commerce and health track closely across all three countries through 2016–2020, with deviations from a common slope remaining small throughout the pre-treatment window, and these two disciplines provide the strongest support for the parallel trends assumption. Creative arts also shows close co-movement, though the small absolute enrolment numbers in NZ produce more volatility in the indexed series.

Education shows a mild but consistent upward divergence of AUS from the controls from 2017 onward, with Australian enrolments growing at a faster rate than both the UK and NZ in the pre-treatment period. While not large in magnitude, this pre-existing divergence is in the same direction as the positive post-treatment DiD estimate, which is consistent with the sign reversal observed when country-specific linear trends are added to the main specification. It suggests the education result may partly reflect a continuation of a pre-existing Australian trend rather than a discrete post-2021 response to the fee change. Engineering shows a similar pattern of AUS outpacing the controls from 2018, introducing analogous concerns for that discipline's estimate. Architecture is the most clearly parallel through 2019 but shows AUS enrolments plateauing while UK and NZ continue growing into 2020, which complicates the post-treatment counterfactual. The pre-treatment window is limited to 2019–2020 for the five short-panel disciplines, which is insufficient to meaningfully assess the parallel trends assumption for those fields.

Data Limitations

Raw Data

UK subject classifications changed in 2019/2020, rendering pre-2019 UK data irreconcilable with later figures for five disciplines. As a consequence, these disciplines enter the panel only from 2019, leaving just one pre-policy observation prior to the 2021 JRGS implementation. This makes it difficult to verify the parallel trends assumption for those disciplines and results in materially higher uncertainty in their DiD estimates relative to the six disciplines with the full 2016–2024 panel. Additionally, more specific sub-disciplines were aggregated into broader category groupings to harmonise data across countries; for example, management and commerce encompasses accounting, marketing, and finance, all of which were subject to the same funding changes. While this does not introduce inconsistencies, it reduces the specificity of the findings. The analysis is also constrained by a limited number of pre- and post-treatment observations, as enrolment data prior to 2016 was unavailable and the post-2021 window is inherently bounded by the current date. Finally, the JRGS applied only to domestic students, yet it was not possible to disaggregate domestic from international enrolments in the Australian source data, meaning the observed totals capture both groups. Since international students were unaffected by the commonwealth funding changes, their inclusion may attenuate the estimated treatment effects.

Identification

The small number of control countries limits the analysis's resilience to country-specific shocks. With only the UK and NZ as comparators, any idiosyncratic shock to either control country in the post-treatment period could bias the DiD estimate. The same concern applies to the treatment side: with Australia as the sole treated unit, any Australia-specific time shock unrelated to the JRGS cannot be separated from the treatment effect. COVID-19 also had an asymmetric effect across countries; Australia's international border closures ran from 2020 to 2022, a longer period than either NZ or the UK, which may have differentially affected Australian enrolment numbers through reduced international student intake during the transition into the post-treatment window. Finally, while NZ provides a second control country and strengthens identification, its enrolment levels are substantially smaller than both Australia and the UK, which may give it a disproportionate influence on the pooled counterfactual despite its relatively small scale.

Robustness and Inference

With only two primary control country clusters ($G=2$, or $G=3$ if Canada is included), cluster-robust standard errors are not feasible: the minimum number of clusters required for reliable cluster inference is approximately 42. HC3 heteroscedasticity-robust standard errors are used instead, justified by the Moulton factor. Permutation inference similarly cannot achieve conventional significance thresholds at this cluster size; with $G=3$ assignments, the minimum achievable permutation p-value is $1/3 \approx 0.33$. The permutation results serve as a model-free diagnostic rather than a formal significance test: for management and commerce, both placebo country assignments produce larger absolute DiD coefficients than the true Australian assignment (permutation $p=1.00$), while for education the true assignment ranks second (permutation $p=0.67$). The most substantive identification concern is that the education parallel trends assumption is not robust to country-specific linear trends: when country-specific trend terms are added to the main specification, the education DiD coefficient reverses sign from $+0.171$ to -0.043 , suggesting that Australian education enrolments were on a diverging trajectory relative to the control countries prior to 2021, which undermines the reliability of that estimate.

Results

Main Results

Table 1 reports the TWFE DiD coefficient (β_3 on did = treated $\times$ post) for all eleven disciplines, estimated with HC3 standard errors.

Table 1. Effect of the Job-Ready Graduates Scheme on University Enrolments

Dependent variable: $\log(\text{annual enrolments})$, Australia vs UK + New Zealand panel. Estimating equation: $\log(\text{enrolments}) \sim \text{treated} + \text{nz_dummy} + \text{did} + C(\text{year})$, HC3 SEs.

Panel A — Priority disciplines (JRG fee reduction)

	Education	Eng. & RT	Health	IT	Nat. Sci.	Environment
DiD (β_3)	+0.171* (0.081)	+0.019 (0.061)	-0.011 (0.039)	+0.141 (0.160)	+0.009 (0.022)	-0.068 (0.083)
Approx. %	+18.6%	+1.9%	-1.1%	+15.1%	+0.9%	-6.5%
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N	27	27	27	27	18	18
Adj. R ²	0.997	0.998	0.999	0.968	1.000	0.999

Panel B — Discouraged disciplines (JRG fee increase)

	Mgmt & C	Soc. & Cult.
DiD (β_3)	-0.243† (0.127)	+0.008 (0.037)
Approx. %	-21.6%	+0.8%
Year fixed effects	Yes	Yes
Country fixed effects	Yes	Yes
N	27	18
Adj. R ²	0.979	0.999

Panel C — Non-priority disciplines

	Arch. & Build.	Creative Arts	Others
DiD (β_3)	-0.054 (0.033)	+0.026 (0.035)	-0.268 (0.241)
Approx. %	-5.2%	+2.6%	-23.5%
Year fixed effects	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes
N	27	27	18
Adj. R ²	0.999	0.998	0.993

Notes: HC3 heteroskedasticity-robust standard errors in parentheses. *** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$ † $p < 0.10$. Country fixed effects absorbed via *treated* (AUS=1) and *nz_dummy* (NZ=1); year fixed effects absorb common shocks including COVID-19. N=27 for 2016–2024 panel; N=18 for 2019–2024 panel (five disciplines). Approx. % = $(\exp(\beta) - 1) \times 100$. Data: AUS DESE; UK HESA; NZ TEC.

Table 1 reports the DiD estimates for all eleven disciplines. The JRG fee reform is associated with a meaningful enrolment response in two disciplines, Education and Management & Commerce, and no detectable effect in the remaining nine.

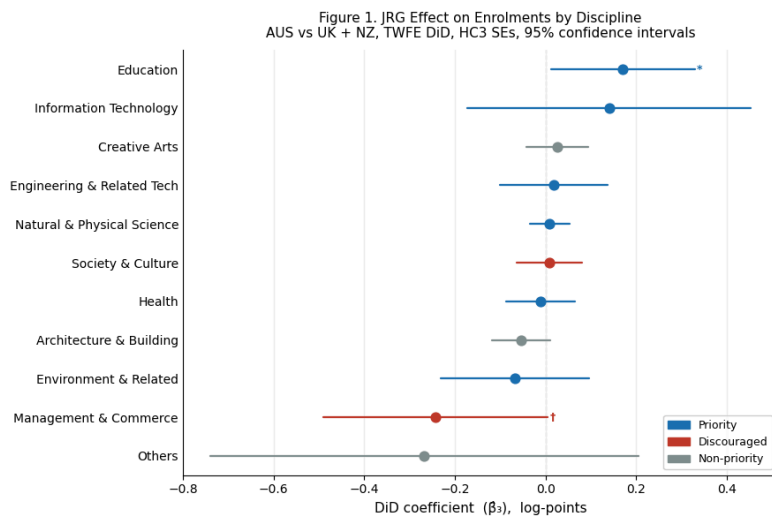
Education (Panel A), where student fees were reduced by approximately 70%, shows a positive and statistically significant DiD coefficient: $\beta_3 = +0.171$ (SE = 0.081, $p = 0.035$), implying approximately 18.6% higher Australian enrolments relative to the UK and New Zealand trend following the 2021 reform. The result survives most robustness checks but should be treated as an upper bound: UK Education enrolments were independently declining post-2021 due to domestic teacher training policy changes, and the coefficient reverses sign when country-specific linear trends are controlled for.

Management & Commerce (Panel B), where fees approximately doubled, shows the opposite: $\beta_3 = -0.243$ (SE = 0.127, $p = 0.055$), implying a 21.6% relative decline in Australian enrolments against trend. This result is only marginally significant and should be treated as suggestive rather than cleanly identified: both wrong-group placebos register significant post-2021 movements in the control countries, and the coefficient collapses to insignificance when two UK pre-treatment observations are dropped.

The remaining nine disciplines produce coefficients that are uniformly small and statistically insignificant (all $p \geq 0.10$). The two disciplines whose fee structures changed most dramatically moved in the expected direction; fields with smaller or no fee changes did not. However, the robustness evidence tempers a strong causal reading: post-2021 global dynamics in international student demand and pre-existing country-specific trends make clean attribution difficult, particularly for Management & Commerce.

Coefficient Plot

Figure 1 plots the same estimates sorted by effect size. Priority disciplines (fee reduction) are shown in blue, discouraged disciplines (fee increase) in red, and non-priority disciplines in grey.



Education stands out clearly among Priority fields. The five other Priority disciplines (Health, Engineering, IT, Environment, Natural Sciences) cluster near zero, while Education alone produces an estimate with its interval lying largely above zero. The Education fee cut, at approximately 70% the most dramatic among Priority fields, produced the clearest enrolment signal.

Among Discouraged disciplines, Management & Commerce is the only field with a notable negative estimate. Society & Culture shows a coefficient close to zero, suggesting the JRG's fee increase had a more pronounced effect on M&C than on its counterpart Discouraged field, though the M&C result carries the identification caveats described above.

The null results across nine disciplines are credible rather than simply imprecise: confidence intervals are narrow and centred near zero. The Education and M&C estimates are of broadly similar absolute size (approximately 0.17 and 0.24 log-points respectively), consistent with roughly symmetric responsiveness to fee changes in opposite directions.

Summary

The JRG is associated with an approximately 18.6% relative increase in Australian Education enrolments above the UK and New Zealand trend following 2021 ($\beta = +0.171$, $SE = 0.081$, $p = 0.035$) and a 21.6% relative decline in Management & Commerce enrolments ($\beta = -0.243$, $SE = 0.127$, $p = 0.055$). Nine other disciplines, including Priority fields such as Health and Engineering that received fee reductions, show no detectable response.

A clean causal attribution is not warranted for either headline result. For Education, the UK experienced an independent post-2021 enrolment decline due to domestic teacher training policy changes, making the estimate an upper bound on the pure JRG effect. For Management & Commerce, global post-COVID international student reallocation produced similar post-2021 declines in the control countries, and the result is sensitive to early UK pre-treatment data points.

The results are consistent with the JRG fee structure shaping enrolment behaviour at the field level, but should be interpreted with caution given a three-country panel, concurrent global demand shifts, and the parallel-trends assumption being untestable with only two comparison countries.

Robustness

Table R1 in the Robustness Checks notebook presents the full battery of alternative specifications. The two headline estimates respond differently under scrutiny.

The Education coefficient is broadly stable across sample restrictions. Excluding 2020 alone or both 2020 and 2021 strengthens rather than weakens the estimate ($\beta = +0.188$ and $+0.218$ respectively), indicating the result does not depend on the COVID baseline year. The sign is preserved under alternative control sets, the UK-only and NZ-only jackknife specifications, and the levels outcome. Three high-leverage observations are present (Cook's $D > 4/N$), and re-estimating after their removal yields $\beta = +0.133$ ($p = 0.064$), directionally preserved but weaker. Two checks raise more fundamental concerns. First, the UK wrong-group placebo is statistically significant ($\beta = -0.172$, $p = 0.035$): UK Education enrolments were themselves declining post-2021, contaminating the control group and inflating the estimated AUS effect. Second, the coefficient reverses sign when country-specific linear trends are added ($\beta = -0.043$), indicating that part of the estimated discontinuity may reflect a pre-existing divergence between Australian and UK enrolment trajectories rather than a policy break.

The Management & Commerce result is less stable. Excluding COVID years strengthens the estimate ($\beta = -0.271$ with 2020 dropped), and restricting to the UK as the sole comparator produces a notably larger coefficient ($\beta = -0.501$, $p < 0.001$), but the NZ-only specification yields $\beta = +0.015$ (effectively zero), revealing that the pooled result is driven almost entirely by the AUS–UK comparison. More critically, both wrong-group placebos are significant: the UK placebo produces $\beta = +0.508$ ($p < 0.001$) and the NZ placebo $\beta = -0.265$ ($p = 0.045$), indicating that the design generates large post-2021 breaks regardless of which country is assigned to treatment. Dropping two UK pre-treatment observations (2016–2017, Cook's $D > 4/N$) causes the coefficient to collapse from -0.243 to -0.161 ($p = 0.181$). The exact permutation test confirms the pattern: with $G = 3$ country clusters, both placebo assignments produce larger absolute coefficients than the real AUS assignment, placing it last among all three possible treatment allocations (permutation $p = 1.00$).

Taken together, the robustness evidence supports a cautious reading of Education and substantially weakens the M&C claim. The Education result survives most specification checks but is bounded above by the UK's concurrent independent decline; the M&C result is consistent with the JRG fee increase but cannot be credibly separated from the global post-2021 reallocation of international students across anglophone systems.

Discussion

The results present a picture that neither fully validates nor fully dismisses the prior literature on JRGS effectiveness. Daly and Lewis (2020) characterised the scheme as a policy unlikely to shift enrolment patterns, a prediction consistent with the broader consensus that tuition price changes exert an inelastic effect on higher education demand at the market level (Langelett et al., 2015; Havranek et al., 2018). Nine of eleven disciplines produce null results, which is largely in line with that prediction. Two fields depart from it: education enrolments rose approximately 18.6% relative to the control countries and management and commerce fell approximately 21.6%, suggesting the JRGS may have produced real effects, but only under specific conditions that the inelastic demand model does not capture on its own.

The concentration of effects in the two disciplines with the largest absolute fee changes is consistent with the elasticity mechanism proposed by Carter and Curry (2011), which locates price responsiveness in higher education not at the aggregate level but between competing alternatives. The JRGS effectively reproduced the conditions of inter-institutional price competition within institutions, across disciplines. Before the reform, maximum student contributions were broadly compressed between 6,500 and 11,000 across all fields. After 2021 that range widened to 4,000–15,000, with education falling conspicuously below the prior floor and management and commerce rising conspicuously above the prior ceiling. Engineering, health, IT, natural sciences, and environment all received fee cuts of between 20% and 45%, reducing their contributions to 7,000–10,000 but keeping them within the previously familiar range, and none produced a statistically significant enrolment response. This pattern is consistent with the existence of a salience threshold: fee changes that keep a discipline within the bandwidth of what students have historically perceived as the normal cost of tertiary study may not register as a meaningful signal in the enrolment decision, while changes that move a field outside that range entirely are more likely to.

The absence of effects in health, engineering, and information technology is particularly striking given that these fields combine the most substantial fee reductions with the most acute documented labour market shortages; health shortage rates reached approximately 80% of Skill Level 1 occupations by 2023 and engineering peaked at 77% in the same year. If the JRGS was designed to increase graduate supply in fields where the labour market signal is strongest, these are precisely the disciplines where the policy should have shown the clearest results. Their null outcomes align with the university enrolment motivation literature: entry into vocational professions such as nursing, medicine, and engineering is shaped by career commitment, prior subject investment, and interpersonal factors that fee changes are unlikely to override (Janke, 2020). The JRGS fee reductions for these fields were also deferred in practice via the HECS-HELP scheme, reducing the immediate financial salience of the change at the point of enrolment decisions. Students not already drawn to those fields by non-financial motivations are unlikely to be redirected toward them by a reduction in deferred annual contribution, suggesting the fee cuts functioned primarily as a transfer to students who would have enrolled regardless rather than as an incentive that expanded the pool of graduates the labour market actually needed.

The robustness evidence substantially constrains the interpretation of both headline results. For education, the most important challenge is the UK wrong-group placebo: UK education enrolments were independently declining post-2021 due to domestic teacher training policy changes, meaning some portion of the estimated AUS–UK divergence reflects movement in the control group rather than a genuine Australian increase. When country-specific linear trends are incorporated into the specification, the education coefficient reverses sign from +0.171 to –0.043, a pattern corroborated by the pre-treatment trend data showing Australian education enrolments already growing faster than the controls from 2017 onward. The most conservative reading of the education result is therefore an upper bound on any genuine JRGS effect, with the true effect possibly smaller and not cleanly separable from a pre-existing trajectory. The management and commerce result faces a more fundamental structural problem: the NZ-only estimate collapses to +0.015, revealing that the pooled result is driven almost entirely by the AUS–UK comparison. With both wrong-group placebos producing statistically significant post-2021 movements and the permutation test placing the true Australian assignment last among all three possible treatment allocations (permutation $p=1.00$), the M&C estimate cannot be credibly isolated from whatever broader post-2021 dynamics were differentially affecting management enrolments across anglophone systems. It is better read as a descriptive finding consistent with the policy's direction than as an identified causal claim.

Evaluated against the JRGS's stated purpose of increasing graduate supply in high-priority, shortage-affected disciplines and reducing enrolment in fields with surplus supply, the results suggest the policy achieved its enrolment objectives in two disciplines but not the ones that matter most for the labour market problem it was designed to address. Education is a genuine shortage field and the observed increase, if real, is consistent with the policy's intent. Management and commerce is a field with persistently low shortage rates, and a decline in enrolments from a surplus-supply discipline is directionally appropriate. The null results across the remaining nine disciplines, however, include the fields with the most severe and sustained shortages: health, where shortage rates have remained above 70% of relevant occupations since 2022, and engineering and IT, which peaked above 75% before partially easing. A policy that succeeded in shifting students away from a low-shortage management field while failing to increase the supply of nurses, engineers, and IT graduates falls well short of the labour market correction it was presented as achieving. This is broadly consistent with the Daly and Lewis (2020) critique, though the evidence here points to a more specific failure mode: the JRGS appears to have been effective enough to produce visible enrolment shifts only at the extremes of its price structure, but insufficiently powerful to overcome the vocational, intrinsic, and structural factors that govern entry into the high-shortage professions where its effects were most needed.

Conclusion

This paper evaluated whether the Job-Ready Graduates Scheme produced enrolment shifts consistent with its intended pricing incentives, using a difference-in-differences estimator with UK and New Zealand enrolments as control groups across eleven disciplines from 2016 to 2024. The headline result is that the JRGS produced statistically significant effects in two of eleven disciplines: education enrolments increased by approximately 18.6% relative to the control countries following the 2021 reform, and management and commerce fell by approximately 21.6%. The remaining nine disciplines, including those combining the most acute labour market shortages with the most substantial fee reductions, produced null results.

Both significant results are consistent with a salience threshold interpretation of the Carter and Curry (2011) mechanism: the JRGS appears to have influenced enrolments only where fee changes moved disciplines outside the historical range of student contribution norms, below the prior floor in education and above the prior ceiling in management and commerce. Disciplines repositioned within the previously familiar bandwidth produced no detectable response. Both headline estimates are, however, materially constrained by robustness evidence. The education coefficient reverses sign under country-specific linear trends and is challenged by a significant UK wrong-group placebo attributable to independent domestic policy changes; the estimate is best treated as an upper bound rather than a point estimate of the JRGS effect. The management and commerce estimate collapses to near zero under NZ-only identification and is ranked last in permutation inference, indicating it cannot be credibly separated from broader post-2021 dynamics common to anglophone higher education systems.

Evaluated against the JRGS's stated objectives, the findings are broadly consistent with the Daly and Lewis (2020) characterisation of the scheme as unlikely to achieve meaningful enrolment reorientation, though the evidence here identifies a more specific failure mode. The policy appears to have produced visible enrolment shifts only at the extremes of its price structure while failing to generate the graduate supply response where the labour market need was greatest. Health, engineering, and information technology registered no significant enrolment change despite receiving the largest proportional fee reductions and exhibiting sustained shortage rates above 70% of relevant Skill Level 1 occupations. Entry into these vocational professions is governed by career commitment, prior subject investment, and intrinsic motivation that fee changes operating through a deferred repayment scheme are unlikely to overcome (Janke, 2020), suggesting the reductions functioned primarily as transfers to students who would have enrolled regardless rather than as supply-expanding incentives.

The policy implication is that price signals alone are insufficient to redirect graduate supply toward shortage-affected disciplines when enrolment decisions are governed primarily by non-financial factors and the financial cost itself is deferred to post-graduation income thresholds. Future evaluations would benefit from longer post-treatment panels as the 2021 cohort matures toward graduation, individual-level data capable of distinguishing compositional from headcount effects, and analysis of completion and employment outcomes to determine whether any demand shifts that did occur translated into the graduate supply the scheme intended to produce.